

EXAM – SOLUTION

*This is a 3 hours exam. Class slides and any handwritten material are allowed.*

YOU MUST WRITE LEGIBLY.

I – TRUE, FALSE, UNCERTAIN? (*1/2 of the points*)

For each of those statements, say whether it is true, false, or uncertain and explain why. You should target around 10 to 15 lines per question. Copying the slides will not bring any points.

1 – It is optimal that unemployment rate is zero.

FALSE. That would be true in a competitive and frictionless labor market: unemployment would correspond to unexploited gains from trade. Things are quite different if there are frictions to trade. If we take the example of matching models, the number of hirings is typically  $H = m(U, V)$  where  $U$  is the number of unemployed and  $V$  the number of vacancies. In such a case,  $U$  can be seen as an input to produce hirings. If the economy needs at each date a flow of hirings (for example because jobs are destroyed), then it is suboptimal to have  $U = 0$ , as it would lead to zero hirings.

2 – In the Ramsey model, the transversality condition states that the level of capital should tend to zero.

FALSE. Let's assume that the economy terminates in period  $T$ . In such an economy, it is optimal for the planner to leave zero capital stock at the end of period  $T$ . Why? Assume that the optimal path leaves a positive amount of capital at the end of period  $T$ . Allocate this capital stock to consumption of period  $T$ . The intertemporal utility will be higher. Hence a contradiction with the assumption that the path was optimal. When one considers an infinite horizon problem, the condition is:

$$\lim_{T \rightarrow \infty} e^{-\rho T} \mu_T \cdot K_T = 0.$$

where  $\mu$  is the shadow value of capital. The condition is not that the *level* of capital should tend to zero, but its discounted value, which is a weaker condition.

3 – Labour-augmenting technical progress is a necessary condition for long run growth of income per capita.

FALSE. We represent technological progress as an expansion of the production set: more output can be produced with the same inputs. Take a “Neo-classical” economy, whose key ingredients are (i) a production function  $Y(t) = F(K(t), L(t))$ , (ii) an equation for capital accumulation  $\dot{K}(t) = Y(t) - C(t)$  where  $Y$  = output,  $L$  = labour,  $K$  = capital, and  $C$  = consumption and where  $\forall t$   $L(t)$  is equal to the workforce which is an exogenous variable.

Technical progress can be modelled in three ways: (1) labour-augmenting technical progress:  $Y(t) = F(K(t), A(t)L(t))$ , (2) capital-augmenting technical progress:  $Y(t) = F(A(t)K(t), L(t))$  and (3) output-augmenting technical progress:  $Y(t) = A(t)F(K(t), L(t))$ .

In the three cases, there will be increasing income per capita. But only in the labor-augmenting case does the model satisfies the five following stylized facts of balanced growth, which are (SF1)  $Y/L$  and  $K/L$  go up over time, (SF2)  $K/Y$  is constant, (SF3) wages go up, (SF4) the profit rate is constant, (SF5) The share of labour and capital in GDP are constant.

4 – Intellectual property rights have ambiguous effects on growth and welfare.

TRUE. On the one hand, in R&D models, innovators typically do not entirely appropriate the social surplus from their innovation. They get profits but part of the surplus goes to the consumers.

This tends to make innovation too low compared with the optimum. Better enforcement of intellectual property rights foster R&D, therefore growth, and is a good thing for welfare. But on the other hand, ideas are non rival goods. The monopoly distortion in price-setting that is created by intellectual property rights tends to push prices above marginal cost. In equilibrium this means that real wages are too low relative to the social optimum, which is bad for welfare.

5 – The speed of convergence is high in the AK model.

FALSE. Consider a Solow model (assume that  $Y$  and  $K$  are per capita):  $Y = AK^\alpha$ ,  $\dot{K} = sY - \delta K$ .  $A$  grows at a constant rate  $g_A$ , both  $Y$  and  $K$  grow at the common constant rate  $g$  in the long run. Let  $\hat{y} = \frac{Y - Y_{LR}}{Y_{LR}}$  be the relative deviation from the BGP (same definition for  $\hat{k}$ ).

By definition, the speed of convergence is given by  $v = -\frac{1}{\hat{y}} \frac{d\hat{y}}{dt}$ .

It can be shown that  $v = (g + \delta)(1 - \alpha)$ .

In an AK model,  $\alpha = 1$ , so that speed of convergence is actually null. This is because there is no decreasing returns in  $K$ : starting with a low  $K$  does not imply that marginal productivity of capital is very high and therefore growth temporarily higher.

6 – Time is continuous and the interest rate is  $r$ . Consider a blue asset (value  $V_b$ ) that pays  $b$  per unit of time and a yellow asset (value  $V_y$ ) that pays  $y$  per unit of time. A green asset is an asset that is blue and that can turn yellow according to a Poisson distribution with intensity  $\alpha$ . Using a discrete time approximation and taking the limit when the length of the period tends to zero, one can prove that

$$rV_g(t) = b + \alpha (V_b(t) - V_y(t)) + \dot{V}(t)$$

FALSE. Consider a discrete time version of the problem. By arbitrage:

$$V_g(t)(1 + r \cdot dt) = b \cdot dt + (1 - \alpha dt)V_g(t + dt) + \alpha dt V_y(t + dt)$$

First order Taylor expansion:

$$V_x(t + dt) = V_x(t) + \dot{V}_x(t)dt$$

so that the arbitrage equation writes

$$V_g(t) + rV_g(t)dt = b \cdot dt + (1 - \alpha dt)(V_g(t) + \dot{V}_g(t)dt) + \alpha dt(V_y(t) + \dot{V}_y(t)dt)$$

Neglecting the terms in  $(dt)^2$  gives

$$rV_g(t) = b + \alpha (V_y(t) - V_g(t)) + \dot{V}_g(t)$$

7 – Unemployment benefits increase unemployment because they shift upwards the Beveridge curve.

FALSE. The basic building block of the Beveridge curve is the matching function, which relates hirings per unit of time to the two key inputs in the search process, unemployment and vacancies:  $H_t = m(U_t, V_t)$ . Here  $H_t$  = the gross hiring rate per unit of time,  $U_t$  = the number of unemployed workers,  $V_t$  = the number of vacant jobs. The matching function is similar to a production function, and we assume it has the same properties including constant returns to scale.

Assume that a fraction  $s$  of all jobs is destroyed per unit of time. Let  $\bar{L}$  = the total labor force,  $L_t$  = employment at  $t$ . Then we can define the hiring, unemployment, and vacancy rates in relation to the total workforce:  $u_t = \frac{U_t}{\bar{L}} = \frac{\bar{L} - L_t}{\bar{L}}$ ,  $v_t = \frac{V_t}{\bar{L}}$  and  $h_t = \frac{H_t}{\bar{L}}$ . Because of constant returns to scale, we can write  $h_t = m(u_t, v_t)$ .

The evolution of the unemployment rate is  $\frac{du}{dt} = -h_t + s(1 - u_t) = -m(u_t, v_t) + s(1 - u_t)$ . The "Beveridge curve" is the locus  $du/dt = 0$  locus in the  $(u, v)$ . It does not depend on unemployment benefits. It is a resource constraint, that does not assume any optimal labor demand nor supply.

8 – With Nash-bargaining, the surplus of a match is efficiently shared.

TRUE. Any sharing rule that does not leave some of the surplus undistributed is efficient. This is the case of the Nash-Bargaining. What can be inefficient is vacancy creation (and job search or job destruction when they are endogenous)

9 – Hours worked per person cannot grow along a balanced growth path.

TRUE. Along a BGP, all variables grow at a constant (positive) rate or do not grow. Hours worked per person are bounded above by the total amount of time available per period. They cannot grow at a constant positive rate without violating the time constraint. Therefore, they cannot grow.

10 – In the real business cycle model, stabilizing output following a productivity shock is Pareto improving.

In the canonical RBC model, allocations are Pareto optimal. Therefore, fluctuations are the socially optimal responses to shocks. No policy can improve welfare. This does not mean that fluctuations are not costly: agents would prefer to live in an economy without shocks to productivity (assuming that they have zero mean).

## II – ENDOGENOUS GROWTH (*1/2 of the points*)

Time is continuous. We use the notation  $X_t = X(t)$ . Consider an economy in which population at time  $t$  is  $L_t$  and grows at the constant rate  $n$ , such that  $\dot{L}_t = nL_t$ . All agents have preferences given by

$$\int_0^\infty e^{-\rho t} \frac{C_t^{1-\theta} - 1}{1-\theta} dt,$$

where  $C$  is consumption defined over the final good of the economy. This good is produced as

$$Y_t = \left[ \int_0^N y_t(i)^\beta di \right]^{\frac{1}{\beta}},$$

where  $y(i)$  is intermediate good  $i$  and  $0 < \beta < 1$ . The production function of each intermediate good is

$$y_t(i) = \ell_t(i),$$

where  $\ell(i)$  is labor allocated to good  $i$ . The wage is denoted  $w_t$ .

New goods are produced by allocating workers to the R&D process, with the production function

$$\dot{N}_t = \eta N_t^\phi \times L_{Rt},$$

where  $\phi \leq 1$  and  $L_{Rt}$  is labor allocated to R&D. Labor market clearing requires

$$\underbrace{\int_0^N \ell_t(i) di}_{L_{Pt}} + L_{Rt} = L_t.$$

$L_{Pt}$  is the amount of labor required for production.

Risk-neutral firms hire workers for R&D. A firm who discovers a new good becomes the monopoly supplier, with a perfectly and indefinitely enforced patent. There is free entry on the R&D market.

**For the time being, let's consider the case  $\phi = 1$  and  $n = 0$ .**

1 – Write the instantaneous profit of the final good firm. Show that the demand for intermediate good  $i$  is

$$y_t(i) = p_t(i)^{\frac{1}{\beta-1}} Y_t.$$

- 2 – Show that profit maximization of each intermediate firm results in them setting price as a constant percentage markup over their marginal cost.
- 3 – Show then that demand for good  $i$  can be written as function of  $Y$  and  $w$ .
- 4 – Using the previous result and the production function of final good, determine the equilibrium wage as a function of  $N_t$ .
- 5 – Using the demand for good  $i$ , express  $L_{Pt}$  as a function of  $Y$  and  $N$ .
- 6 – Show that the profit of an intermediate firm can be written as a function of  $Y$  and  $w$ .
- 7 – We now turn to the R&D sector. Let  $Z_t$  denote the present discounted value of the profits of an intermediate good producer at time  $t$ . Next consider a firm undertaking R&D to create a new intermediary. Explain why does the free entry condition on the R&D market writes

$$Z_t \dot{N}_t - w_t L_{Rt} = 0.$$

- 8 – Derive an expression for  $Z$  as a function of  $N$ . (Hint: use the law of motion of  $N$ )
- 9 – Let us restrict to balanced growth path (BGP) and guess that  $N$  grows at a constant rate  $x$  (needed to be verified at some point). Using that guess and the expression for  $Z$ , derive the growth rate of  $Z$  along a BGP.
- 10 – Why does the the following equation holds? What does it mean? ( $r$  is the interest rate,  $\pi_t$  is the instantaneous profit of any intermediate firm at a symmetric equilibrium where all intermediate firms are taking the same decisions; this is the equilibrium that we consider here)

$$rZ_t = \pi_t + \dot{Z}_t.$$

- 11 – Using all the previous equations, derive an expression for  $Y$  as a function of  $r, x, N_0$  and  $t$ . Express the growth rate of  $Y$  (and hence  $C$ ) as a function of  $\beta$  and  $x$ . Denote this equation (1).
- 12 – Without doing the algebra, explain how do we get the following relation from the household intertemporal maximization problem:

$$\frac{\dot{C}_t}{C_t} = \frac{1}{\theta}(r - \rho). \quad (2)$$

How do you interpret this relation?

- 13 – Use (1) and (2) to obtain a relation (3) between  $x$  and  $r$ . Comment.
- 14 – We now need to determine  $x$  in equilibrium. Use the R&D technology and (3) to derive a equation between  $L_R$  and  $r$ . Call the relation ( $\mathcal{A}$ ).
- 15 – Using the expression of  $L_P$  of question 5 and the one of  $Y$  in question 11, get an expression of  $L_P$  as a function of  $r$  and  $x$ . Using  $L_R = L - L_P$ , obtain a second relation ( $\mathcal{B}$ ) between  $L_R$  and  $x$ .
- 16 – Compute the equilibrium level of  $L_R$ . Check that  $x$  is indeed constant and derive its equilibrium level. Compute also the equilibrium level of  $r$ .
- 17 – Compute the growth rate of the economy (which is the growth rate of  $Y$  or  $C$ ). How does it depend on  $\theta$ ? Why?
- 18 – How does the growth rate depend on  $L$ ? Why? Discuss the empirical relevance of that result.

**19 – We now want to characterize the balanced growth path equilibrium when  $\phi < 1$  and  $n > 0$ .** Explain how to derive the new free entry condition

$$Z_t \eta N_t^\phi L_{Rt} - w_t L_{Rt} = 0.$$

**20 –** Derive an expression for  $Z_t$  as a function of  $N_t$ .

**21 –** Using  $rZ_t = \pi_t + \dot{Z}_t$  and the expression of the profit, derive the growth rate of  $Y$  as a function of  $x$ .

**22 –** Derive the growth rate of  $L_{Pt}$  as a function of  $x$ .

**23 –** Derive the growth rate of  $Y$  assuming that the economy began on a balanced growth path at date 0.

**24 –** Does the growth rate of the economy depend on  $L$ ? Why? Discuss.

# ΕΞΑΓΓ - ΠΑΚΕΤΟ 1 Π2

2013-2014

## SOLUTION

$$\underline{\text{Exam}} : M = \int_0^{\infty} e^{-\rho t} \frac{C_t^{1-\theta}}{1-\theta} dt$$

$$Y_t = \left( \int_0^{N_t} y_i(t)^\beta di \right)^{1/\beta}$$

$$y_{it} = l_{it}$$

$$\dot{N}_t = \eta N_t^\varphi L_{Rt}$$

$$L_t = L_{Rt} + \underbrace{\int_0^{N_t} l_{it} di}_{L_{Pt}}$$

$$\dot{L}_t = n L_t$$

- Assume  $\phi = 1$  and  $\alpha = 0$ .

1) Profit:  $\pi_{it} = p_{it} y_{it} - w_t l_{it}$   
with  $l_{it} = y_{it}$

$$\Rightarrow \pi_{it} = (p_{it} - w_t) y_{it}$$

Demand:  $\max u = \left( \int_0^{N_t} y_{it}^\beta di \right)^{1/\beta}$

$$s.t. \int_0^N p_i y_i di = Y_t \quad (1)$$

Foc:  $\frac{1}{y_i} \cdot \frac{1}{\beta} \times \beta y_i^{\beta-1} u^{\frac{1}{\beta-1}} = \lambda p_i$

Ratio of two Foc

$$\left( \frac{y_i}{y_j} \right)^{\beta-1} = \frac{p_i}{p_j}$$

$$y_i = \left( \frac{p_i}{p_j} \right)^{\frac{1}{\beta-1}} y_j$$

$$p_i y_i = p_i^{\frac{1}{\beta-1}+1} p_j^{\frac{1}{1-\beta}} y_j \quad ; \quad \frac{1}{\beta-1}+1 = \frac{\beta}{\beta-1}$$

↳ use the BC:

$$\int_0^N p_i y_i di = p_j^{\frac{1}{1-\beta}} y_j \int_0^N p_i^{\frac{\beta}{\beta-1}} di = Y$$

$$\Rightarrow y_j = p_j^{\frac{-1}{1-\beta}} \left( \int_0^N p_i^{\frac{\beta}{\beta-1}} di \right)^{-1} Y = \boxed{\left( \frac{p_j}{P} \right)^{\frac{1}{1-\beta}} \frac{Y}{P} = y_j}$$

$$\Rightarrow P^{-1} + \frac{1}{1-\beta} = P \frac{\beta}{1-\beta} = \left[ \int p_i^{\frac{\beta}{1-\beta}} d\omega_i \right]$$

$$\Rightarrow \underline{P} = \left[ \int p_i^{\frac{\beta}{1-\beta}} d\omega_i \right]^{\frac{1-\beta}{\beta}}$$

↳ one can normalize  $\omega$  to 1 (numerical)

2) pb of an intermediate firm:

$$\begin{aligned} \max \pi_f(i) &= y_f(i) (p(i) - w_v) \\ &= p_i(i)^{\frac{1}{\beta-1}} (p_i(i) - w) \end{aligned}$$

$$\begin{aligned} \text{FOC}/p_i(i): \frac{1}{\beta-1} p_i^{\frac{1}{\beta-1}-1} (p_i - w) + p_i^{\frac{1}{\beta-1}} &= 0 \\ \Leftrightarrow \frac{1}{\beta-1} \left( 1 - \frac{w}{p_i} \right) &= -1 \Rightarrow 1-\beta = -\frac{w}{p_i} \end{aligned}$$

$$1-\beta = 1 - \frac{w}{p_i} \Leftrightarrow \frac{w}{p_i} = \beta \Rightarrow \boxed{p_i = \frac{w}{\beta}}$$

3) then  $y_i = p(i)^{\frac{1}{\beta-1}} y$

$$\boxed{y_i = \left( \frac{w}{\beta} \right)^{\frac{1}{\beta-1}} y}$$

4)  $y_f = \left[ \int_0^N y_i^{\beta} d\omega_i \right]^{\frac{1}{\beta}} = \left[ N \left( \frac{w}{\beta} \right)^{\frac{\beta}{\beta-1}} y^{\beta} \right]^{\frac{1}{\beta}}$

$$\Leftrightarrow 1 = N^{\frac{1}{\beta}} \left( \frac{w}{\beta} \right)^{\frac{\beta}{\beta-1}} \Leftrightarrow \boxed{w = \beta N^{\frac{1-\beta}{\beta}}}$$



$$\begin{aligned}
 5] \quad L^p &= \int_0^N y_i di = N_t \left( \frac{w}{p} \right)^{\frac{1}{\beta-1}} y \\
 &= N \left( \frac{\beta N^{\frac{1-\beta}{\beta}}}{\beta} \right)^{\frac{1}{\beta-1}} y \\
 &= N N^{-\frac{1}{\beta}} y = N^{\frac{\beta-1}{\beta}} y = N^{\frac{\beta-1}{\beta}} y \\
 &\boxed{L_{Pr} = N^{-\frac{1-\beta}{\beta}} y}
 \end{aligned}$$

$$\begin{aligned}
 6] \quad \pi(i) &= y(i) (p(i) - w) = p_i^{\frac{1}{\beta-1}} y (p_i - w) \\
 &= \left( \frac{w}{p} \right)^{\frac{1}{\beta-1}} y \left( \frac{w}{p} - w \right)
 \end{aligned}$$

$$\begin{aligned}
 \pi(i) &= \left( \frac{w}{p} \right)^{\frac{1}{\beta-1} + 1} y (1 - \beta) \\
 \Rightarrow \quad \pi(i) &= \left( \frac{w}{p} \right)^{-\frac{\beta}{1-\beta}} (1 - \beta) y
 \end{aligned}$$

$$\begin{aligned}
 7] \quad \text{Cost of R\&D: } w_t L_{Rt} \\
 \text{with } L_{Rt}, \text{ one obtains } \dot{N}, \text{ whose value is } Z_t \\
 \rightarrow \text{ free entry } \boxed{Z_t \dot{N}_t - w_t L_{Rt} = 0}
 \end{aligned}$$

8] Use  $\dot{N} = \gamma N L_R$

$$Z \dot{N} - w L_R = 0 \Leftrightarrow Z \gamma N L_R - w L_R = 0$$

$$\Leftrightarrow Z = \frac{w}{\gamma N} \quad \text{and} \quad w = \beta N^{\frac{1-\beta}{\beta}}$$

$$\Rightarrow Z = \frac{\beta}{\gamma} N^{\frac{1-\beta}{\beta}-1}$$

$$\Rightarrow \boxed{Z_t = \left(\frac{\beta}{\gamma}\right) N_t^{\frac{1-2\beta}{\beta}}}$$

9] BGP  $\frac{\dot{N}}{N} = \alpha \Rightarrow N_t = N_0 e^{\alpha t}$

$$Z_t = \left(\frac{\beta}{\gamma}\right) N_0^{\frac{1-2\beta}{\beta}} e^{\alpha \left(\frac{1-2\beta}{\beta}\right) t}$$

$$\hookrightarrow \boxed{\frac{\dot{Z}}{Z} = \alpha \frac{(1-2\beta)}{\beta}}$$

10]  $\alpha Z = \pi + \dot{Z}$

$\hookrightarrow$  arbitrage by a parent: get  $\pi + \dot{Z}$   
result value

$\hookrightarrow$  put money in the bank: get  $\alpha Z$

11

$$n - \dot{Z} = \pi$$

$$\frac{1-2\beta}{\beta} + 1 = \frac{1-\beta}{\beta}$$

$$\text{Use } \begin{cases} \pi = (1-\beta) y \left(\frac{w}{\beta}\right)^{\frac{\beta}{1-\beta}} \\ w = \beta N^{\frac{1-\beta}{\beta}} \\ = \left(\frac{\beta}{h}\right) N_0^{\frac{1-2\beta}{\beta}} e^{\alpha \left(\frac{1-2\beta}{\beta}\right) t} \end{cases}$$

$$\frac{\dot{Z}}{Z} = \alpha \left(\frac{1-2\beta}{\beta}\right) \Rightarrow \dot{Z} = \alpha \left(\frac{1-2\beta}{\beta}\right) Z$$

$$\left(\frac{\beta}{h}\right) N_0^{\frac{1-2\beta}{\beta}} e^{\alpha \left(\frac{1-2\beta}{\beta}\right) t} \left(2 - \alpha \frac{(1-2\beta)}{\beta}\right) = (1-\beta) y N^{-1}$$

$$\Rightarrow \boxed{y = \left(\frac{\beta}{h}\right) \frac{1}{1-\beta} \left(2 - \alpha \frac{(1-2\beta)}{\beta}\right) N_0^{\frac{1-\beta}{\beta}} e^{\alpha \left(\frac{1-\beta}{\beta}\right) t}} \left(N_0 e^{\alpha t}\right)^{-1}$$

and therefore

$$\boxed{\frac{\dot{y}}{y} = \alpha \frac{1-\beta}{\beta}} \quad (1)$$

12) Households max

$$\int_0^{\infty} e^{-\rho t} \frac{C_t^{1-\theta}}{1-\theta} dt$$

$$st \quad \dot{W}_t = n W_t + \underbrace{Y_t}_{income} - C_t$$

$$+ \lim_{t \rightarrow \infty} W_t e^{-\rho t} = 0 \quad (\text{transversality condition})$$

$$\text{For } C_t^{-\theta} e^{-\rho t} = \lambda e^{-\int_0^t \rho dv}$$

$$\text{take log derivative, eliminate } \lambda \Rightarrow \boxed{\frac{\dot{C}}{C} = \frac{1}{\theta} (n - \rho)} \quad (2)$$

interpretation:

if  $n > \rho$ , agents save  $\Rightarrow$  then  $C$  grows  $\Rightarrow \frac{\dot{C}}{C} > 0$

13) we have in eq  $Y = C$

$$(1) (2) \Rightarrow \boxed{\frac{1}{\theta} (n - \rho) = \alpha \left( \frac{1-\beta}{1-\beta} \right)} \quad (3)$$

it is the modified golden rule

- high growth rate  $\alpha \Rightarrow$  need high  $n$  to induce savings
- if  $\theta$  small, lot of intertemporal substit: a lower  $n$  is needed to induce high growth  $\alpha$

14)

$$R \& D \text{ techs: } \dot{N} = \eta N L_R \Rightarrow \frac{\dot{N}}{N} = \pi = \eta L_R$$

$$(3) \quad \frac{1}{\theta} (2 - \rho) = \pi \left( \frac{1 - \beta}{\beta} \right)$$

$$\Rightarrow \boxed{\frac{1}{\theta} (2 - \rho) = \eta L_R \left( \frac{1 - \beta}{\beta} \right)} \quad (A)$$

15)

$$q5: L_{Pr} = N^{\frac{\beta-1}{\beta}} \gamma$$

$$q11: \gamma = \left( \frac{1}{1-\beta} \right) \left( 2 - \pi \left( \frac{1-2\beta}{\beta} \right) \right) \left( \frac{\beta}{h} \right) \underbrace{N^{\frac{1-\beta}{\beta}} e^{\pi \left( \frac{1-\beta}{\beta} \right) t}}_{N^{\frac{1-\beta}{\beta}}}$$

$$\hookrightarrow \gamma N^{\frac{\beta-1}{\beta}} = \boxed{L_P = \left( \frac{1}{1-\beta} \right) \left( 2 - \pi \left( \frac{1-2\beta}{\beta} \right) \right) \left( \frac{\beta}{h} \right)} \quad N^{\frac{1-\beta}{\beta}}$$

$$L_R = L - L_P$$

$$\pi = \eta L_R$$

$$\Rightarrow \boxed{L_R = L - \frac{1}{1-\beta} \left( 2 - \eta L_R \left( \frac{1-2\beta}{\beta} \right) \right) \left( \frac{\beta}{h} \right)} \quad (B)$$

16) solve (B), (A)

$$(A): \pi = \theta \eta \left( \frac{1-\beta}{\beta} \right) L_R + \rho$$

$$\hookrightarrow (B): L_R = L - \frac{1}{1-\beta} \left( \frac{\beta}{h} \right) \left[ \rho + \left( \theta \eta \frac{1-\beta}{\beta} - \eta \left( \frac{1-2\beta}{\beta} \right) \right) L_R \right]$$

Solve:

$$L_R = \frac{(1-\beta)(L - \frac{p\beta}{\gamma})}{\beta + \theta(1-\beta)}$$

$$n = \frac{2\gamma\theta(1-\beta)^2 + p\beta^2(1 + (1-\beta)\theta)}{\beta(\beta + \theta(1-\beta))}$$

17)  $\frac{\dot{N}}{N} = n = \gamma L_R = \frac{\gamma(1-\beta)(L - \frac{p\beta}{\gamma})}{\beta + \theta(1-\beta)} = \frac{(1-\beta)(\gamma L - p\beta)}{\beta + \theta(1-\beta)}$

Finally  $\frac{\dot{Y}}{Y} = \frac{1-\beta}{\beta} n$

$$\Rightarrow \frac{\dot{Y}}{Y} = \frac{(1-\beta)^2(\gamma L - p\beta)}{\beta(\beta + \theta(1-\beta))} \quad \text{growth rate of the economy}$$

The growth rate is lower when  $\theta$  is high.

why? : high  $\theta \Rightarrow$  low intertemporal substitution  
 $\Rightarrow$  need high interest rate to induce savings and growth  $\Rightarrow$  low value of future discounted profits  
 $\Rightarrow$  low  $Z \Rightarrow$  low R&D  $\Rightarrow$  low growth.

- 12) growth is high when  $L$  is high  
 $\Rightarrow$  scale effect of population on growth  
 $\rightarrow$  this seems not to be observed in the data

Assume now  $\phi < 1, n > 0$

$$\frac{\dot{L}}{L} = n, \quad \dot{N} = \eta N^\phi L_R$$

13)  $\left\{ \begin{array}{l} \text{cost of hiring } L_R: w L_R \\ \text{gain: } Z \times \eta N^\phi L_R \end{array} \right.$

$\hookrightarrow$  zero profit:  $Z \eta N^\phi L_R - w L_R = 0$

all the calculations from 1) to 7) are the same. In particular,  $w = \beta N^{\frac{1-\beta}{\beta}}$

20

Plug in the zero profit condition

$$Z_t = \left(\frac{\beta}{\gamma}\right) N^{\frac{1-\beta}{\beta} - \phi}$$

Assume  $\frac{\dot{N}}{N} = \alpha$  (as before) to obtain

$$\frac{\dot{Z}}{Z} = \alpha \left( \frac{1-\beta-\beta\phi}{\beta} \right)$$

21

Use again  $nZ = \pi + \dot{Z}$  (as before) to obtain

$$\frac{\dot{Y}}{Y} = \alpha \left( \frac{1}{\beta} - \phi \right)$$

22

We have (as before)

$$L_{P,t} = Y_t N_t^{\frac{\beta-1}{\beta}}$$

$$\Rightarrow \frac{\dot{L}_P}{L_P} = \frac{\dot{Y}}{Y} + \left( \frac{\beta-1}{\beta} \right) \frac{\dot{N}}{N}$$

$$= \alpha \left( \frac{1}{\beta} - \phi \right) + \frac{\beta-1}{\beta} \alpha$$

$$\frac{\dot{L}_P}{L_P} = \alpha (1 - \phi)$$



23) We have  $L = L_p + L_R$

so that  $\dot{N} = n N^q (L - L_p)$  (\*)

If the economy starts on a BGP:

$$N_t = N_0 e^{nt}$$

$$L_t = L_0 e^{nt}$$

$$L_p = L_{p0} e^{n(1-q)t}$$

so that (\*)  $\Leftrightarrow N_0 n e^{nt} = n N_0 e^{qnt} (L_0 e^{nt} - L_{p0} e^{n(1-q)t})$

For a BGP to exist, we need  $L$  and  $L_p$  to grow at the same rate

We therefore need  $n = n(1-q)$

$$\Leftrightarrow \boxed{n = \frac{n}{1-q}}$$

and therefore, given that  $\frac{\dot{y}}{y} = n(1-q)$

$$\boxed{\frac{\dot{y}}{y} = \left(\frac{n}{1-q}\right)(1-q)}$$

24) Now the growth rate of the economy does not depend on  $L$ , but only on  $\frac{\dot{L}}{L} = n$ .

Why? because there are now decreasing returns to scale in  $L$  in the R&D technology (and not constant returns as before).